# A Formal Model for a Fully Homomorphic Encryption Scheme Based on Group Rings

## 1 Key Generation

The key generation phase establishes the private key $sk = \{a, b, z, t\}$. First, a modulus $M$ is chosen such that the plaintext space forms the ring $\mathbb{Z}_M$. A group $G$ with a subgroup $H \subseteq G$ is defined, where $H$ should exhibit a maximally large number of automorphisms, $|Aut(H)|$, while the group size $n = |H|$ is kept small. Additionally, a homomorphism $s : G \to H$ is defined, which can only be computed efficiently with knowledge of the trapdoor $t$. The structure of $G$ requires that the group operation (addition) is efficient to perform, whereas inversion (and thus subtraction) must be computationally hard.

### 1.1 Calculation of Parameters and Formal Grouping

To compute $z$, an initial random value $r \in \mathbb{Z}_M[H]$ is chosen. Subsequently, the parameter $a = r^k$ is determined, where $k = |\mathbb{Z}_M^*|$ corresponds to the order of the multiplicative group of $\mathbb{Z}_M$.

To formalize linear decryption and the inner product, we define the projection onto the basis of group elements. An element $x \in \mathbb{Z}_M[H]$ can be uniquely represented as a linear combination $x = \sum_{h \in H} x_h h$ with coefficients $x_h \in \mathbb{Z}_M$. We define the vector-isomorphism $\pi : \mathbb{Z}_M[H] \to \mathbb{Z}_M^{|H|}$, which groups a ring element according to the group elements of $H$ and extracts the corresponding coefficient vector:

$$\pi(x) = (x_{h_1}, x_{h_2}, \ldots, x_{h_n})^T \quad \text{for } H = \{h_1, \ldots, h_n\}$$

Building on this, a system of linear equations (SLE) of the form $Az = 1$ is solved. Here, $A_i = \pi(a^i)$ is the matrix of coefficients, exactly sorted and grouped by the variables of the group $H$.

If the SLE is not solvable for the vector $z = (z_{h_1}, \ldots, z_{h_n})^T \in \mathbb{Z}_M^{|H|}$, a new random $r$ is generated. Once a solution is found, $z$ defines the linear decryption homomorphism $f : \mathbb{Z}_M[H] \to \mathbb{Z}_M$ via the formal inner product of the grouped coefficients:

$$f(x) = \langle \pi(x), z \rangle = \sum_{i=1}^{n} x_{h_i} z_{h_i}$$

### 1.2 Calculation of $b$ and Evaluation of $f(ab)$

The parameter $b$ is computed as $b = a^x - a^y$ for $x \neq y$. The essential property $f(ab) = 0$ can be proven using the linearity of the inner product as follows:

$$f(ab) = f(a(a^x - a^y)) = f(a^{x+1} - a^{y+1})$$

By applying the linear function $f$ and utilizing the property of the SLE—where by construction $f(a^i) = \langle \pi(a^i), z \rangle = 1$ holds for all relevant powers $i$—we obtain:

$$f(a^{x+1} - a^{y+1}) = f(a^{x+1}) - f(a^{y+1}) = 1 - 1 = 0$$

This guarantees that the term $b$ is completely annihilated when evaluated by $f$ during the decryption process.

# 2 Encryption and Homomorphic Properties

Encryption takes place within the ring $\mathbb{Z}_M[G]$. An auxiliary function is defined:

$$Enc'(m, sk, g) = ma + kb$$

for a random value $k \in \mathbb{Z}_M$, where additionally every variable of the resulting polynomial is multiplied from the right by a group element $g \in G$.

The final ciphertext is designed as an algebraic fraction:

$$c = Enc(m, sk) = [Enc'(mr, sk, g), Enc'(r, sk, g)]$$

for a random element $g \in G$ as well as a random scalar $r \in \mathbb{Z}_M$.

Homomorphic operations (addition and subtraction) are performed analogously to classical fraction rules. An essential advantage of this scheme is that the ciphertext does not accumulate any additional components after a multiplication. A valid ciphertext $c \in \mathbb{Z}_M[G]$ is constructed such that all group elements contained within it, $var(c)$, lie strictly within a single left coset of $H$:

$$\exists x \in G \text{ such that } var(c) \subset xH$$

Since the product of two elements from (possibly different) cosets falls back into a well-defined coset, the dimensional structure of the ciphertext remains invariant under homomorphic multiplications. The vector space does not expand.

## 2.1 Proof of Homomorphism (Addition and Multiplication)

The homomorphic properties are derived directly from the algebraic fraction rules and the linearity as well as the multiplicativity of the evaluation function $Dec'$ on the plaintext space. Let $c_1 = [N_1, D_1]$ and $c_2 = [N_2, D_2]$ be two ciphertexts, where $N_i = Enc'(m_i r_i)$ represents the numerator and $D_i = Enc'(r_i)$ represents the denominator. By definition, $Dec'(N_i) = m_i r_i$ and $Dec'(D_i) = r_i$.

For homomorphic addition, cross-multiplication is used:

$$c_{add} = c_1 \oplus c_2 = [N_1 D_2 + N_2 D_1, D_1 D_2]$$

Decryption yields the correctly added plaintext:

$$Dec(c_{add}) = \frac{Dec'(N_1 D_2 + N_2 D_1)}{Dec'(D_1 D_2)} = \frac{m_1 r_1 r_2 + m_2 r_2 r_1}{r_1 r_2} = m_1 + m_2$$

For homomorphic multiplication, numerators and denominators are multiplied componentwise:

$$c_{mul} = c_1 \otimes c_2 = [N_1 N_2, D_1 D_2]$$

Here too, decryption yields the correct product of the plaintexts:

$$Dec(c_{mul}) = \frac{Dec'(N_1 N_2)}{Dec'(D_1 D_2)} = \frac{m_1 r_1 \cdot m_2 r_2}{r_1 \cdot r_2} = m_1 \cdot m_2$$

### 2.2 Absence of Cross-Terms during Homomorphic Evaluation

Homomorphic operations on the ciphertexts follow classical fraction rules. Let two ciphertexts $c_1 = [\alpha_1, \beta_1]$ and $c_2 = [\alpha_2, \beta_2]$ be given, which represent the fractions $\frac{\alpha_1}{\beta_1}$ and $\frac{\alpha_2}{\beta_2}$. Homomorphic addition is defined as:

$$c_{add} = [\alpha_1\beta_2 + \alpha_2\beta_1, \beta_1\beta_2]$$

Homomorphic multiplication is defined as:

$$c_{mul} = [\alpha_1\alpha_2, \beta_1\beta_2]$$

A fundamental problem with traditional homomorphic schemes is the explosion of ciphertext size due to cross-terms during multiplication. In this scheme, this is prevented by utilizing cosets in the group ring $\mathbb{Z}_M[G]$. A valid ciphertext may only contain variables from a specific coset of $H$:

$$C' = \{v \in \mathbb{Z}_M[G] \mid var(v) \subset xH \text{ for some } x \in G\}$$

To demonstrate that no irreducible cross-terms arise, we consider the multiplication of two polynomials $v_1 \in xH$ and $v_2 \in yH$. Every element in $v_1$ has the form $xh_i$ and in $v_2$ the form $yh_j$ (with $h_i, h_j \in H$). Multiplying these polynomials formally creates cross-terms:

$$v_1 \cdot v_2 = \left(\sum_i m_i(xh_i)\right) \cdot \left(\sum_j m'_j(yh_j)\right) = \sum_{i,j}(m_i m'_j) \cdot (xh_i \cdot yh_j)$$

Because $(xh_i \cdot yh_j)$ falls back into the unique coset $(xy)H$ due to group properties, all these cross-terms collapse within the ring $\mathbb{Z}_M[G]$. Instead of creating $|H|^2$ isolated variables, the coefficients $m_i m'_j$ that map to the same basis element in $(xy)H$ are simply added modulo $M$. The resulting number of basis variables (components) after a multiplication therefore never exceeds $|H| = n$. The same applies to addition given the same coset. Consequently, the dimensionality of the ciphertext remains strictly compact.

## 3 Decryption

For a fractional ciphertext $c = [c_1, c_2]$, the plaintext $m$ is reconstructed via the division of the separately evaluated components:

$$m = Dec(c, sk) = \frac{Dec'(c_1, sk)}{Dec'(c_2, sk)}$$

The decryption function of the individual components is:

$$Dec'(c_i, sk) = f(c_i a)$$

Since the variables of $c_i$ lie in $G$ (or $H$), but the function $f$ is defined on $\mathbb{Z}_M[H]$, all group elements must first be projected into the subgroup $H$ using the homomorphism $s : G \to H$ and the private trapdoor key $t$ prior to evaluating $f$.

### 3.1 Proof of Homomorphism via Linear Algebra

To understand why $Dec'$ is homomorphic, we look at the interaction between the variables $a$ and $b$. A (simplified) ciphertext has the form $c = ma + kb$. Substituting this into the function $Dec'$ yields:

$$\begin{aligned} Dec'(c, sk) &= f(ca) \\ &= f((ma + kb)a) = f(ma^2 + kba) \\ &= m \cdot f(a^2) + k \cdot f(ab) \end{aligned}$$

From the construction of the SLE $Az = 1$, we know that $f(a^i) = 1$ for all powers $i > 0$, hence $f(a^2) = 1$. At the same time, it was proven that $f(ab) = 0$. The equation thus simplifies to:

$$Dec'(c, sk) = m \cdot 1 + k \cdot 0 = m$$

Due to this linear algebra property, $a$ acts as an evaluator that annihilates the random mask $kb$ (since $b$ lies in the kernel of the functional induced by $a$) while simultaneously leaving the message component $m$ invariant. The linearity of the functional $f$ therefore guarantees that homomorphic additions and multiplications (while preserving the coset structure) pass directly through to the messages $m$.

**Prevention of Cross-Terms via $ab = 0$**
Examining the algebraic structure of ciphertexts in the form $c = ma+rb$ reveals that homomorphic operations do not generate unwanted cross-terms with respect to the function $f$.

When adding two ciphertexts $c_1 = m_1a + r_1b$ and $c_2 = m_2a + r_2b$, the components are simply combined linearly:

$$c_1 + c_2 = (m_1a + r_1b) + (m_2a + r_2b) = (m_1 + m_2)a + (r_1 + r_2)b$$

Due to linearity, no cross-terms naturally arise here.

During multiplication, expanding the product initially yields mixed terms:

$$c_1 \cdot c_2 = (m_1a + r_1b) \cdot (m_2a + r_2b) = m_1m_2a^2 + m_1r_2ab + r_1m_2ba + r_1r_2b^2$$

However, because the fundamental property $ab = 0$ applies in this system, the mixed cross-terms ($ab$ and $ba$) immediately annihilate each other. The product thus collapses directly and reduces to its essential components without the ciphertext expanding due to new, irreducible terms:

$$c_1 \cdot c_2 = m_1m_2a^2 + r_1r_2b^2 = (m_1m_2)a + (r_1r_2)b$$

## 4 Security and Resistance against Structural Attacks

The security of the system is based on the algebraic indistinguishability of the cosets and the computational hardness of inversion within the group $G$.

A potential attack consists of linearly decrypting the SLE. To do this, an attacker would need to map multiple ciphertexts $c^{(1)}, c^{(2)}, \dots$ onto exactly the same automorphism $\phi \in Aut(H)$ to obtain compatible systems of equations. Finding this mapping would require the attacker to analyze the internal structure of a ciphertext's variables.

### 4.1 The Difference Matrix and the Inversion Problem

Let $c$ be a ciphertext with variables $V = \{v_1, \ldots, v_n\} \subset xH$. To characterize the structure up to isomorphism and match permutations, an attacker could attempt to construct a relative difference matrix $D \in G^{n \times n}$:

$$D_{i,j} = v_i - v_j = (x + h_i) - (x + h_j) = h_i - h_j$$

(Written here in additive notation to clarify the reference to the difference). The matrix $D$ characterizes the variable vector up to the constant offset $x$ and is invariant to translations, even in commutative groups.

However, the construction of $D$ strictly requires calculating the difference $v_i - v_j$, which algebraically corresponds to addition with the inverse: $v_i + (-v_j)$. According to the definition of the group $G$, while addition is easy, the inversion $-v_j$ (or $v_j^{-1}$ in multiplicative notation) is a computationally hard problem (*Hardness Assumption: Inversion is hard, thus subtraction is hard*). Without the ability to compute the inverses of the variables, the attacker is incapable of constructing the matrix $D$.

**Example (Loss of security if inversion were possible):**
Assume the inversion in $G$ were trivially computable. For a ciphertext with the variables $v_1 = x + h_1$ and $v_2 = x + h_2$ (where $x$ is the unknown offset of the coset), an attacker could form the difference:

$$v_1 - v_2 = (x + h_1) - (x + h_2) = h_1 - h_2$$

The result $h_1 - h_2$ is completely independent of the secret offset $x$. By pairwise comparison of these deterministic difference patterns between two ciphertexts $c^{(1)}$ and $c^{(2)}$, one could definitively verify whether they are structurally isomorphic. This allows sorting the variables and mapping them directly to the same automorphism, thereby breaking the SLE linearly.

### 4.2 Protection via the Size of the Automorphism Group

Without the matrix $D$, the attacker cannot construct a permutation matrix $\Pi$ for which $D^{(1)} = \Pi D^{(2)} \Pi^T$ holds. Consequently, it is impossible to deliberately align different ciphertexts algebraically such that they operate under the same automorphism.

The attacker is thus forced to rely on purely random collisions. The probability $\Pr[\text{Collision}]$ that two randomly chosen ciphertexts fall into the same orbit of the automorphism group is given by:

$$\Pr[\text{Collision}] = \frac{1}{|Aut(H)|}$$

Since the subgroup $H$ is explicitly chosen so that $|Aut(H)|$ is extremely large, this probability is negligible.

**Example (Loss of security with few automorphisms):**
If the number of automorphisms were very small, e.g., $|Aut(H)| = 2$. If an attacker intercepts merely 3 random ciphertexts, the Pigeonhole Principle guarantees that at least two of these ciphertexts must necessarily be subject to the same automorphism. The attacker would simply have to try the two possible mappings until the resulting SLE becomes consistently solvable for the secret vector $z$.

### 4.3 Identification of Automorphisms via the Cayley Table

To illustrate the theoretical threat potential, consider how an attacker might find an automorphism if the structure of $H$ were completely exposed. An automorphism is a bijective

mapping of the group onto itself that exactly preserves the group structure: $\phi(a \circ b) = \phi(a) \circ \phi(b)$. If the Cayley table (the complete multiplication table) of the group is known, an automorphism can be derived by permuting the elements:

1. First, the generators of the group are identified based on the cyclic patterns in the table.
2. A valid automorphism must strictly map generators to generators of the same order.
3. By systematically swapping the rows and columns (corresponding to the mapping candidates for the generators), a transformed table is generated.
4. If the newly generated table perfectly structurally matches the original Cayley table, the chosen mapping represents a valid automorphism.

This procedure underscores the necessity of designing the system such that inversion remains hard. Only then is it prevented that the attacker can structurally prepare ciphertexts and successfully apply such mappings.

Without aligning the mappings onto the same automorphism, the linear relations extractable from the ciphertexts remain mutually incompatible and linearly independent concerning the secret vector $z$. Establishing a global, overdetermined system of equations to extract $z$ is thereby mathematically thwarted.